CHAPTER 2

LITERATURE REVIEW

Large language models (LLMs) are prone to producing fluent but factually incorrect output, a direct consequence of their autoregressive generation (Ji *et al.*, 2023). This unreliability is especially problematic for knowledge-intensive question answering, where reasoning must be verifiable against external facts. Recent work addresses this by grounding models in knowledge graphs, but keeping generation strictly aligned with a graph without sacrificing fluency remains an open challenge.

This chapter reviews the connections between LLM hallucination, multi-step validation, and structured grounding, assessing mitigation strategies, from prompting to constrained decoding, to identify where soft-grounding methods fall short and where this project's technical gaps lie.

2.1 Large Language Models and the Hallucination Problem

Modern LLMs use decoder-only Transformer architectures with causal self-attention (Vaswani *et al.*, 2017); Llama-3.1-8B (Dubey *et al.*, 2024) is a prominent example and the backbone used in recent KG-constrained decoding work (Luo *et al.*, 2025). Generation is autoregressive: given vocabulary V, input x, and prior tokens $y_{(<t)}$, the model computes $P(y_t|y<t, x)$ over the full vocabulary at each step, so every token retains a non-zero probability of being generated regardless of correctness. Pretraining on next-token prediction over vast unlabelled text yields powerful downstream capabilities (Brown *et al.*, 2020; Touvron, Martin, and Stone, 2023; Dubey *et al.*, 2024), but the resulting knowledge is static and statistical, with no mechanism to verify outputs against dynamic external facts.

Definition and Structural Causes.
Hallucination is fluent, coherent content that is factually unsupported or contradictory (Ji *et al.*, 2023), split into *intrinsic* (contradicts source context) and *extrinsic* (unverifiable) types (Huang *et al.*, 2025). Reported rates range from 15–60% across tasks (Ji *et al.*, 2023), with roughly one-third of reasoning steps hallucinated even under direct KG access (Luo *et al.*, 2025). The cause is structural: generation has no external verification component, errors propagate forward as conditioning context for subsequent steps, and the full vocabulary remains reachable at every step regardless of correctness (Brown *et al.*, 2020; Huang *et al.*, 2025; Ji *et al.*, 2023).

The Scaling Paradox.
Larger models do not resolve hallucination. TruthfulQA shows some categories exhibit *inverse scaling*, where larger models reproduce incorrect beliefs more effectively (Lin, Hilton, and Evans, 2022); RLHF training favors expectation-aligned over verifiable responses (Perez *et al.*, 2023); scale helps frequent entity combinations but not compositional, unseen-combination questions , the structure of multi-hop KGQA (Kandpal *et al.*, 2023); and tail-entity hallucination persists in GPT-4 regardless of size (Mallen *et al.*, 2023). Hallucination is thus a structural feature of autoregressive generation, not a scaling deficiency.

2.2 Chain-of-Thought and Its Limits

Chain-of-Thought Prompting.
Chain-of-thought (CoT) prompting (Bosma *et al.*, 2022) is the dominant paradigm for eliciting multi-step reasoning, using worked examples to teach the model to decompose problems into sub-steps. It yields substantial accuracy gains on arithmetic, commonsense, and multi-hop QA benchmarks, with the capability emerging around 100 billion parameters (Bosma *et al.*, 2022); zero-shot CoT achieves similar gains simply by adding "Let's think step by step" (Kojima *et al.*, 2022). However, CoT has a structural limitation: each intermediate step is generated by the same autoregressive mechanism, so an incorrect step can propagate unchecked, and a chain can remain fluent even while containing fabricated steps.

Extensions: Self-Consistency and Tree-of-Thought.
Self-consistency (Wang *et al.*, 2023) samples multiple independent CoT chains and takes the majority answer, reducing sensitivity to any single hallucinated chain, though a majority of independently hallucinated chains can still yield a wrong answer, especially on deep multi-hop questions. Tree-of-Thought (Cao *et al.*, 2023) treats generation as search through a tree of partial reasoning sequences, using the LLM as both generator and evaluator; pruning still relies on the model's own parameters without external verification.

The Faithfulness Ceiling.
All prompting-based methods few-shot, CoT, self-consistency, tree-of-thought modify the LLM's *input* without changing its *generation mechanism*. Since decoding always occurs over the full vocabulary, every token retains a strictly positive probability at each step. True structural faithfulness requires invalid tokens to receive exactly zero probability, which can only be achieved by directly altering the decoding algorithm itself (Geng *et al.*, 2023; Willard and Louf, 2023).

2.3 Knowledge Graphs and Knowledge Graph Question Answering

Knowledge Graphs.

A knowledge graph is a structured database of verified facts, represented as directed, labelled triplets (e, r, e') where $e, e' \in$ E are entities and $r \in$ R is a relation type. Figure 2.1 illustrates a simplified fragment of such a structure. Freebase (Bollacker *et al.*, 2008), the KG underlying the evaluation benchmarks used in this work, encodes tens of millions of triplets across geography, history, entertainment, and science (Yih **et al.**, 2016; Talmor and Berant, 2018).

[Figure 2.1 A simplified example of a knowledge graph fragment.]

Unlike an unstructured text corpus, every KG triplet is explicitly verified and machine-readable in deterministic form, making KGs suitable as hard constraints on generation: a candidate token can be verified via exact lookup rather than probabilistic matching. A multi-hop reasoning path of length L is a sequence $(e_0, r_1, e_1, \dots, r\_L, e\_L)$ in which each consecutive triplet is a verified member of T (Lan *et al.*, 2022).

2.3.2 Knowledge Graph Question Answering.

KGQA requires identifying the answer entity e_L to a question q by reasoning through a chain of verified KG triplets: given q with question entities E_q ⊆ E identified by entity linking, the system must find the terminal entity of the correct reasoning path in G (Lan *et al.*, 2022). Early approaches relied on semantic parsing or IR over KG subgraphs; more recent work uses LLMs as retrievers, iterative agents, or generators conditioned on retrieved subgraph context (Jiang *et al.*, 2023a; Jin *et al.*, 2024; Luo *et al.*, 2024) improving fluency but reintroducing hallucination risk.

2.3.3 Evaluation Benchmarks.

WebQSP (Yih *et al.*, 2016) contains 4,737 Freebase questions requiring one to two hops, evaluated via Hits@1 and F1. ComplexWebQuestions (CWQ; Talmor and Berant, 2018) extends WebQSP with conjunction, composition, and comparison operations across 34,000 questions requiring up to four hops illustrating a case with many more valid paths, making correct path selection harder.

2.3.4 Multi-Hop Complexity.

For an entity with average out-degree d, the number of valid paths of length L grows as |E_q| × d^L. For well-connected Freebase entities ($d \approx 20$) with up to three question entities, a three-hop question can generate up to 24,000 structurally valid paths, of which only one is correct (He *et al.*, 2021; Lan *et al.*, 2022) meaning the model must select the correct next token from a vast field of valid-but-incorrect alternatives at every step.

2.4 Constrained Decoding for Faithful Text Generation

Constrained decoding restricts generation to a predefined valid set at each step by directly intervening in the decoding mechanism. Unlike prompting, which changes the input to the LLM, constrained decoding alters the token selection process itself. Invalid tokens receive zero probability before sampling, making their generation structurally impossible.

2.4.1 Logit Masking.

The core mechanism behind constrained decoding is logit masking (Geng *et al.*, 2023; Willard and Louf, 2023): a binary mask over the vocabulary computed at each generation step, setting invalid tokens' log-probability to $-\infty$ before the softmax. This forces their selection probability to exactly zero, making them structurally impossible to generate.

2.4.2 Grammar- and Format-Constrained Decoding.

Geng *et al.* (2023) introduce Grammar-Constrained Decoding (GCD), restricting generation to strings conforming to a context-free grammar via an Earley parser oracle. Willard and Louf (2023) reduce the computational cost with a finite-state machine (FSM) formulation that precomputes an index from automaton states to valid vocabulary subsets, enabling amortised $O(1)$ lookup. De Cao *et al.*(2021) introduce GENRE, limiting generation to entity names via a trie of known surface forms. Other constrained decoding methods include Synchromesh (Poesia *et al.* , 2022) for program synthesis, PICARD (Scholak, Schucher, and Bahdanau, 2021) for schema-constrained SQL parsing, NeuroLogic Decoding (Lu *et al.*, 2021) for propositional logic constraints, and grid beam search (Hokamp and Liu, 2017) for lexical constraints; Zhang *et al.*(2023) confirm that finite-state constraints admit efficient enforcement via precomputed indices.

The shared limitation across these methods is that they enforce output *format* rather than *factual content*, a grammatically valid output can still be entirely hallucinated. Extending format constraints to KG path constraints requires an oracle that encodes the knowledge graph's relational structure itself.

2.4.3 Knowledge Graph-Constrained Decoding.

Applying constrained decoding to KG faithfulness uses the KG itself as the constraint oracle: at each generation step, only tokens corresponding to valid continuations of a KG reasoning path are permitted, making hallucination of KG facts structurally impossible.

Graph-Constrained Reasoning (GCR) (Luo *et al.*, 2025) builds a KG-Trie (a prefix tree of all KG paths within L hops of the question entities, constructed via BFS before generation) and uses it as the constraint oracle during beam-search decoding. GCR achieves 92.6 Hits@1 on WebQSP and 75.8 on CWQ with 100% structural faithfulness on a Llama-3.1-8B backbone. The trie is fixed once built: the valid token set does not change based on what has already been generated.

Decoding on Graphs (DoG) (Li *et al.*, 2025) addresses GCR's static oracle with step-wise trie expansion: after each entity is locked in, the constraint trie expands with all of that entity's KG neighbors, making the oracle topologically dynamic. DoG achieves 90.99 Hits@1 on WebQSP and 76.08 on CWQ with 100% structural faithfulness. However, expansion includes all neighbors regardless of relevance to the question, and step-wise construction sacrifices the efficiency of precomputed structures.

RouterKGQA (Yuan *et al.*, 2026) routes between a specialized KG-reasoning model and a general LLM based on question complexity, using SBERT embeddings (nomic-embed-text-v1) to filter candidate answers *after* generation, so semantic scoring operates at answer selection rather than at the token-constraint level, with paths created first and filtered afterward.

2.4.4 Beam Search and Trie-Based Constraints.

Beam Search. First introduced in the HARPY system (Lowerre, 1976), beam search keeps the k most promising partial hypotheses at each step, expanding, ranking by log-probability, and pruning back to the top k (Holtzman *et al.*, 2020) reducing early-error multiplication while remaining computationally viable.

The Prefix Trie. A trie (Fredkin, 1960) is a tree structure where a node's position encodes the prefix it represents and its branches encode valid next tokens. A pre-computed trie yields the valid next-token subset in $O(1)$ time, making it the most scalable option for factual constraints at inference speed (Willard and Louf, 2023).

The KG-Trie. GCR's KG-Trie (Luo *et al.*, 2025) serializes verified multi-hop KG paths into string form mapped to the LLM's token vocabulary, with each edge representing a specific valid token rather than a single character. Merging beam search with a KG-Trie oracle, as shown in Figure 2.2, turns decoding into a strictly bounded traversal of the prefix tree.

[Figure 2.2 Visualization of KG-Trie Constrained Generation.]

At each autoregressive step, beam hypotheses map to specific trie states. The model computes a distribution over its full vocabulary, but a mask forces any token not present as a valid child node to exactly zero probability ($-\infty$ before softmax). Beam search then advances strictly along permissible branches, restricting generation to pre-verified paths while still letting the model express preference among factually valid chains.

2.5 Retrieval and Soft Grounding

Retrieval-augmented generation (RAG) (Lewis *et al.*, 2020) is the leading approach for grounding LLM outputs in external knowledge: a dense retriever selects relevant passages by embedding similarity, and a reader LLM generates the answer from that context. Fusion-in-Decoder (Izacard and Grave, 2021) encodes passages separately and merges them in the decoder. Self-RAG (Asai *et al.*, 2024) adds reflection tokens for when to retrieve and whether passages support claims. FLARE (Jiang *et al.*, 2023b) triggers retrieval when next-token confidence drops. IRCoT (Trivedi *et al.,* 2023) interleaves retrieval with CoT steps, improving multi-hop accuracy. GraphRAG (Edge *et al.*, 2024) builds community-structured KGs from document collections for complex relational queries.

Within KGQA specifically, several approaches enhance RAG with structured context: He *et al.* (2021) retrieve KG subgraphs via attention-guided traversal; Yasunaga *et al.* (2021) combine text passages and KG neighborhoods through graph neural networks; and Shi *et al.* (2021) retrieve KG paths as natural language context strings. Sentence-BERT (Reimers and Gurevych, 2019) encoders, producing dense vectors where cosine similarity proxies semantic alignment, are applied in these systems to rank candidate KG paths as soft retrieval signals (Shi *et al.*, 2021; Saxena, Chakrabarti, and Talukdar, 2022).

2.5.1 The Structural Limitation of Soft Grounding.

Despite empirical gains, all RAG approaches share a structural limitation: retrieved context only *softly* influences generation. As shown in Figure 2.3, the retriever determines what the LLM sees, not what it is permitted to generate. Because the generation mechanism itself is unchanged, every token retains a nonzero probability at every step including tokens corresponding to no verified KG fact. Structural faithfulness with probability one is unreachable under any architecture that modifies only the input context without modifying token selection (Luo *et al.*, 2025; Geng *et al.*, 2023).

[Figure 2.3 — Standard RAG vs. KG-Constrained Decoding]

2.6 Synthesis: Three Unresolved Gaps

This review traces two research paths , constrained decoding for structural faithfulness and semantic similarity for relevance-guided reasoning that have developed largely in parallel, without being combined at the constraint-oracle level. Constrained decoding methods (Luo *et *al.*, 2025; Li *et al.*, 2025) achieve structural faithfulness but build their oracles without considering question semantics or generation context, producing valid token sets containing many irrelevant paths. Semantic similarity methods (Shi *et al.*, 2021; Saxena, Chakrabarti, and Talukdar, 2022; Yuan *et al.*, 2026) improve relevance but act only as soft signals affecting retrieval or post-hoc selection, without changing generation itself.

Three gaps emerge from this intersection:

Gap 1: The Permissiveness Problem. Current KG-constrained decoding frameworks build oracles from graph structure and question entities alone, ignoring question semantics and generation state. For multi-hop questions, this yields valid token sets with thousands of structurally correct but semantically irrelevant paths, forcing the LLM to sort through them using its own knowledge reintroducing the hallucination risk constrained decoding was meant to eliminate (Luo *et al.*, 2025; Li *et al.*, 2025).

Gap 2: The Static-Dynamic Efficiency Tradeoff.
GCR's precomputed static trie is efficient but semantically unaware and unchanging; DoG's step-wise expansion adds dynamism but requires trie reconstruction at every step without semantic filtering (Li *et al.*, 2025; Willard and Louf, 2023). No framework combines progressive, semantically-responsive narrowing of the valid set with the efficiency of precomputed structures.

Gap 3: The Absence of a Constraint Quality Metric.
No existing work measures constraint permissiveness independently of downstream answer accuracy, making it impossible to track progress on reducing irrelevant paths or compare frameworks on constraint quality alone.

These gaps point to a shared challenge: integrating semantic relevance scoring directly into the constraint oracle, conditioning the valid token set jointly on the knowledge graph, question semantics, and evolving generation state.

Table 2.1 summarizes the key works reviewed, categorized by approach, structural faithfulness, and semantic conditioning.